Grade 12 Physics
Gravitational, Electrical, Magnetic Fields Test

First Name: _____

Last Name: _____

Directions:

- Please answer to 2 decimal points
- The test is designed to be completed in 75 minutes

For grading use only

Page:	2	3	4	5	6	7	Total
Points:	6	19	20	10	10	20	85
Score:							

Multiple Choice (10 marks)

1. (1 point) Which of the following is true about gravitational fields?
   A. Gravitational fields only exist near massive objects like planets.
   B. Gravitational fields are regions where a mass experiences a force.
   C. Gravitational fields are strongest at the center of a planet.
   D. Gravitational fields do not depend on the mass of the object creating them.

2. (1 point) What is the direction of the electric field around a positive point charge?
   A. Radially inward.
   B. Radially outward.
   C. Tangential to the charge.
   D. There is no electric field around a positive charge.

3. (1 point) A magnetic field is produced by:
   A. Stationary electric charges.
   B. Moving electric charges.
   C. Gravitational forces.
   D. Static electric fields.

4. (1 point) The force experienced by a charged particle in a magnetic field is maximum when:
   A. The particle is stationary.
   B. The particle moves perpendicular to the magnetic field.
   C. The particle moves parallel to the magnetic field.
   D. The particle is uncharged.

5. (1 point) The gravitational force between two masses is inversely proportional to:
   A. The product of the masses.
   B. The square of the distance between them.
   C. The sum of the masses.
   D. The distance between them.

6. (1 point) The unit of electric field strength is:
   A. Newton (N).
   B. Newton per Coulomb (N/C).
   C. Joule (J).
   D. Volt (V).

7. (1 point) A proton and an electron are placed in the same electric field. Which experiences a greater acceleration?
   - A. The proton.
   - B. The electron.
   - C. Both experience the same acceleration.
   - D. Neither accelerates.

8. (1 point) The magnetic field inside a long solenoid is:
   - A. Zero.
   - B. Uniform and parallel to the axis.
   - C. Non-uniform and radial.
   - D. Strongest at the ends.

9. (1 point) The force between two parallel current-carrying wires is:
   - A. Always attractive.
   - B. Attractive if currents are in the same direction.
   - C. Repulsive if currents are in the same direction.
   - D. Independent of the current direction.

10. (1 point) The work done in moving a charge in an electric field depends on:
    - A. The path taken.
    - B. The potential difference.
    - C. The charge's mass.
    - D. The magnetic field present.

Long Answer (40 marks)

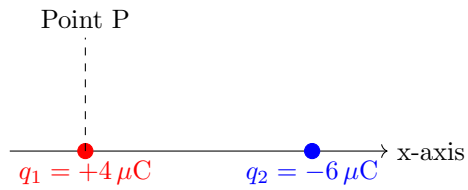
11. A satellite of mass 500 kg orbits the Earth at a height of 300 km above the surface. The radius of the Earth is 6.37×10^6 m, and its mass is 5.97×10^{24} kg.
    (a) (5 points) Calculate the gravitational force acting on the satellite.
    (b) (5 points) Determine the orbital speed of the satellite.
    (c) (5 points) Find the period of the satellite's orbit.

12. A proton moves with a velocity of 2×10^6 m/s perpendicular to a magnetic field of 0.5 T.
    (a) (5 points) Calculate the magnetic force acting on the proton.
    (b) (5 points) Determine the radius of the proton's circular path.

13. Two point charges, $q_1 = 3\,\mu$C and $q_2 = -5\,\mu$C, are placed 10 cm apart.
    (a) (5 points) Calculate the electric force between the charges.
    (b) (5 points) Determine the electric field at the midpoint between the charges.

14. A wire carrying a current of 5 A is placed in a uniform magnetic field of 0.2 T. The wire is 0.5 m long and makes an angle of 30° with the magnetic field.
    (a) (5 points) Calculate the magnetic force acting on the wire.
    (b) (5 points) Determine the direction of the force using the right-hand rule.

15. Two point charges, $q_1 = +4\,\mu\text{C}$ and $q_2 = -6\,\mu\text{C}$, are placed 12 cm apart along the x-axis, as shown below.



A point P is located 5 cm above q_1.

(a) (4 points) Calculate the electric field at point P due to q_1.

(b) (4 points) Calculate the electric field at point P due to q_2.

(c) (2 points) Determine the direction of the **net** electric field at P.

16. A proton enters a uniform magnetic field $\vec{B} = 0.4\,\mathrm{T}\,\hat{k}$ with a velocity $\vec{v} = 3 \times 10^5\,\mathrm{m/s}\,\hat{i} + 4 \times 10^5\,\mathrm{m/s}\,\hat{j}$.
    (a) (4 points) Calculate the magnetic force acting on the proton.
    (b) (3 points) Describe the resulting motion of the proton.
    (c) (3 points) What would happen to the motion if the proton were replaced with an electron?

17. A 500 kg satellite orbits Earth ($M_{\mathrm{Earth}} = 5.97 \times 10^{24}\,\mathrm{kg}, R_{\mathrm{Earth}} = 6.37 \times 10^6\,\mathrm{m}$) at an altitude of 400 km.
    (a) (4 points) Calculate the gravitational potential energy of the satellite.
    (b) (3 points) Determine the satellite's orbital speed.
    (c) (3 points) Explain why the gravitational potential energy is **negative**.